SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	**CSE-AI**	**Date:**	**13-10-26**

Code of Conduct

I ____T.Rajashree(192472393)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___T.Rajashree_____

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

A Total spectrum = 45 MHz = 45,000 kHz

Bandwidth per simplex channel = 25 kHz

Formula:

Full-duplex channels=Total spectrum/2*Simplex bandwidth

Calculation: $N = \frac{45{,}000}{2\times25} = \frac{45{,}000}{50}$

N=900

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

A

Total voice channels = 560

Number of cells in the cluster = 7

Formula:

$Channelspercell = \frac{Totalchannels}{Clustersize}$

Calculation:

$N = \frac{560}{7}$

N=80

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

A:Number of carriers = 100

Time slots per carrier = 8

Carrier spacing = 200 kHz

Formula:

Total voice channels=Carriers*Time slots per carrier

Calculation:

$$N = 100 \times 8$$

N=800

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10·log10 of the ratio).

A:Given: Base station power = 80 W, handset power = 4 W.

$$Ratio = 10\log_{10}\left(\frac{80}{4}\right)$$

10\log_{10}(20)=13.01

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

A:C/I = Carrier power − Interference power

= −60 − (−92) = 32 dB

Answer: 32 dB.

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

A:Number of channels = 42,000 / 300 = 140

Channels per cell = 140 / 7 = 20

Answer: 20 channels per cell.

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

A:Received power = 64 / 2^4 = 64 / 16 = 4 W

Answer: 4 W.

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment

obeying an inverse-square power law. By what factor should transmit power be scaled up?

A:Power factor = $(2R/R)^2 = 4$

Answer: Transmit power must be increased by a factor of 4.

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

A:Total channels = 4 × 15 = 60

Answer: 60 channels.

Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

A:Velocity = 100 × 1000 / 3600 = 27.78 m/s

Doppler shift = vf/c

$= (27.78 \times 900 \times 10^6) / (3 \times 10^8) = 83.3$ Hz

Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

A:Slot duration = 5 / 8 = 0.625 ms

Answer: 0.625 ms.

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

A:Number of carriers = 24,000 / 200 = 120

Answer: 120 carriers.

Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

A:Transmit power = $10 \log_{10} (8000) = 39.03$ dBm

Received power = 39.03 − 33 = 6.03 dBm

Answer: 6.03 dBm.

Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell

A:sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

FSPL = $32.44 + 20 \log_{10}(1900) + 20 \log_{10}(2.5)$

FSPL = 105.97 dB

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

A:Total capacity = 6 × 250 = 1500 calls

Answer: 1500 simultaneous calls.

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

A:Average call duration = 3 × 60 = 180 seconds

Active calls = 4000 × 180 = 720,000

Answer: 720,000

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

A:Bandwidth = 150 × 25 = 3750 kHz

Answer: 3.75 MHz.

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

A:Number of gaps = 30 − 1 = 29

Total guard bandwidth = 29 × 40 = 1160 kHz

Answer: 1.16 MHz.

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

A:Cell diameter = 2 × 1.5 = 3 km

Handoffs = 90 / 3 = 30

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base

A:station serving 250 users. What is the network's total user-handling capacity?

Total users = 8 × 40 × 250 = 80,000

Answer: 80,000 users.

Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

A:Radius = 400 m = 0.4 km

Cell area = $R^2 = 0.4^2 = 0.16$ km²

Number of cells = 400 / 0.16 = 2500

Answer: Approximately 2500 cells.

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

A:Number of channels = 6000 / 30 = 200

Answer: 200 channels.

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

A:Traffic = (1200 × 4 × 2.5) / 60 = 200 Erlangs

Answer: 200 Erlangs.

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

A:Number of users = 40,000 / 400 = 100

Answer: 100 simultaneous users.

Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

A: Total capacity = 1200 × 600 = 720,000

Answer: 720,000 simultaneous users.